

Answer Key — What Is a Force?

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (c) **a push or a pull** — force is a push or a pull resulting from interaction.
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- Q2** (b) **newton (N)** — force is measured in newtons (N).
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- Q3** (c) **shape** — squeezing changes the balloon's shape.
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- Q4** (b) **an interaction between at least two objects** — a force comes from two objects interacting.
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- Q5** (b) **False** — it may have balanced forces that cancel out.
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- Q6** **direction** — force can change direction of motion.
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- Q7** **n** — the newton's symbol is N.
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Short answer

- Q8** A push or pull on an object resulting from its interaction with another object; SI unit = newton (N).
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- Q9** Start motion from rest, change speed, change direction of motion, change shape (any/all).
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- Q10** A force needs at least two objects; when one pushes another, both experience a force, and it ends when they stop interacting.
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Long / application

- Q11** (i) decreases speed / stops it; (ii) makes it move from rest (and gives direction); (iii) changes its direction and speed.
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- Q12** Speed: pushing a cart makes it go faster. Direction: hitting a moving ball with a bat turns it. Shape: pressing a balloon flattens it. Each change is caused by a force.
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